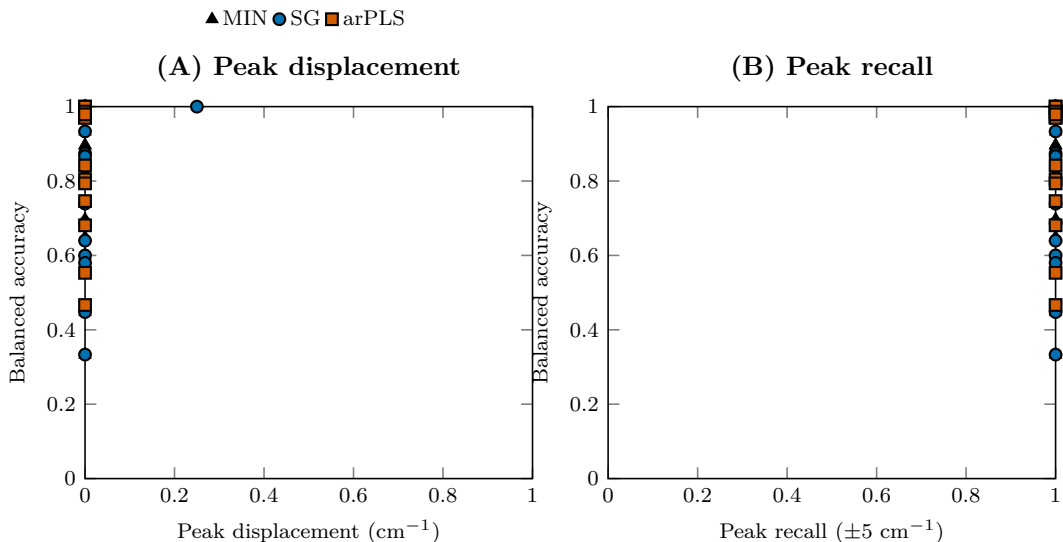




RQ-S05. Each plotted marker is one accepted point: a comparison domain crossed with an action (policy), not an individual spectrum. Panel A shows stored median peak displacement; panel B shows stored median reference-peak recall. Both use balanced accuracy on y. MIN (black triangle), SG (blue circle) and arPLS (vermillion square) are overlaid. MIN uses minimal min-max scaling; SG combines impulse replacement and smoothing; arPLS combines impulse replacement and baseline correction. All use 400-1800 cm<sup>-1</sup> and final [0,1] scaling. Per-axis NA points (A=12, B=12) remain in the table/CSV and are not plotted, so the two axes may carry different plotted denominators. Stored 10th/90th percentiles are summaries, not confidence intervals. P08-F07 joins accepted F02 balanced-accuracy pairs to accepted preservation summaries. The source study comprised 598 original spectra from 69 physical masters across 10 instruments, partitioned into 17 station-instrument domains (13 held comparison domains, 4 exploratory domains). Endpoint M01 uses individual predictions and M06 uses mean model probabilities; neither is a mean spectrum. Model selection and calibration are source-only, and the selected CNN identity is fixed across preprocessing. Peak diagnostics are descriptive against the frozen observed-spectrum reference and are not clean-chemistry truth. No causal distinction, winner selection, or new-instrument guarantee is implied. Summaries are stored-observation-weighted medians and 10th/90th percentiles (linear interpolation), not confidence intervals or master-equal chemical estimates. Original preservation counts and held-prediction support counts are kept separate. The baseline-span proxy is not measured fluorescence, and clipped fraction does not prove physical detector saturation. M06 averages model probabilities within each sample/instrument and then equally across instruments, not input spectra or hard labels.